# Band-1 Hadamard order 13724 – Classical Paley-I (q=13723); IN SPK via Paley-I; Gap 2052 Still OPEN

H_13724 Paley-I PASS; Paley-II FAIL/SKIP (q=6861=3*2287 NOT prime; ===1 mod 4 but composite); Kronecker FAIL/SKIP (4*3431; 3431===3 mod 4 NOT Hadamard-admissible; 3431=47*73 (odd composite)); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-07

**Document class.** Scientific computational / construction certificate (Band-1).

**Program.** Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

**Parents.** `2026-10-07_Band1_Hadamard_13712_PaleyI`; `2026-10-07_Band1_Hadamard_13692_PaleyI`; `2026-10-07_Band1_Hadamard_13688_PaleyI`; `2026-10-06_Band1_Hadamard_13680_PaleyI`; `2026-10-06_Band1_Hadamard_13628_PaleyI`; `2026-10-06_Band1_Hadamard_13620_PaleyI`; `2026-10-06_Band1_Hadamard_13592_PaleyI`; `2026-10-06_Band1_Hadamard_13568_PaleyI`; `2026-10-06_Band1_Hadamard_13524_PaleyI`; `2026-10-06_Band1_Hadamard_13500_PaleyI`; `2026-10-06_Band1_Hadamard_13488_PaleyI`; `2026-10-06_Band1_Hadamard_13464_PaleyI`; `2026-10-06_Band1_Hadamard_13452_PaleyI`; `2026-10-06_Band1_Hadamard_13412_PaleyI`; `2026-10-06_Band1_Hadamard_13400_PaleyI`; `2026-10-06_Band1_Hadamard_13368_PaleyI`; `2026-10-06_Band1_Hadamard_13340_PaleyI`; `2026-10-06_Band1_Hadamard_13332_PaleyI`; `2026-10-06_Band1_Hadamard_13328_PaleyI`; `2026-10-06_Band1_Hadamard_13292_PaleyI`; `2026-10-06_Band1_Hadamard_13268_PaleyI`; `2026-10-06_Band1_Hadamard_13260_PaleyI`; `2026-10-06_Band1_Hadamard_13220_PaleyI`; `2026-10-06_Band1_Hadamard_13188_PaleyI`; `2026-10-06_Band1_Hadamard_13184_PaleyI`; `2026-10-06_Band1_Hadamard_13172_PaleyI`; `2026-10-05_Band1_Hadamard_13164_PaleyI`; `2026-10-05_Band1_Hadamard_13160_PaleyI`; `2026-10-05_Band1_Hadamard_13152_PaleyI`; `2026-10-05_Band1_Hadamard_13148_PaleyI`; `2026-10-05_Band1_Hadamard_13128_PaleyI`; `2026-10-05_Band1_Hadamard_13104_PaleyI`; `2026-10-05_Band1_Hadamard_13100_PaleyI`; `2026-10-05_Band1_Hadamard_13064_PaleyI`; `2026-10-05_Band1_Hadamard_13044_PaleyI`; `2026-10-05_Band1_Hadamard_13008_PaleyI`; `2026-10-05_Band1_Hadamard_13004_PaleyI`; `2026-10-05_Band1_Hadamard_12984_PaleyI`; `2026-10-05_Band1_Hadamard_12980_PaleyI`; `2026-10-05_Band1_Hadamard_12968_PaleyI`; `2026-10-05_Band1_Hadamard_12960_PaleyI`; `2026-10-05_Band1_Hadamard_12924_PaleyI`; `2026-10-05_Band1_Hadamard_12920_PaleyI`; `2026-10-05_Band1_Hadamard_12912_PaleyI`; `2026-10-05_Band1_Hadamard_12908_PaleyI`; `2026-10-04_Band1_Hadamard_12900_PaleyI`; `2026-10-04_Band1_Hadamard_12824_PaleyI`; `2026-10-04_Band1_Hadamard_12800_PaleyI`;

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2026-09-24_Band1_Hadamard_10332_PaleyI; 2026-09-24_Band1_Hadamard_10304_PaleyI;
2026-09-24_Band1_Hadamard_10272_PaleyI; 2026-09-23_Band1_Hadamard_10268_PaleyI;
2026-09-23_Band1_Hadamard_10224_PaleyI; 2026-09-23_Band1_Hadamard_10212_PaleyI;
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2026-09-23_Band1_Hadamard_10152_PaleyI; 2026-09-23_Band1_Hadamard_10140_PaleyI;
2026-09-23_Band1_Hadamard_10112_PaleyI; 2026-09-23_Band1_Hadamard_10104_PaleyI;
2026-09-23_Band1_Hadamard_10100_PaleyI; 2026-09-23_Band1_Hadamard_10068_PaleyI.

**Lab.** `/workspace/experiments/band1/hadamard_13724_paleyI/`.

**Cutoff.** 2026-10-07T11:26:16Z (~7:26AM ET).

**Honesty stamp (non-negotiable).**

This memo claims **finite construction of order 13724 only (verified max|HHˆT-nI|=0), via classical Paley-I q=13723 (prime ===3 mod 4). 13724 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 13724: Paley-I **PASS** ($q = 13723$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=6861=3*2287 NOT prime; ===1 mod 4 but composite); Kronecker **FAIL/SKIP** (note $13724 = 4 \cdot 3431$; $3431 \equiv 3 \pmod 4$ NOT Hadamard-admissible; $3431 = 47 \cdot 73$ (odd composite)). **13724** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **13712+13692+13688+13680+13628+13620+13592+13568+13524+13500+13488+13464+13452+134** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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## Abstract

Finite construction of order **13724**.

- $H_{13724}$ = Paley-I ($q = 13723$) – classical SPK
- Paley-II for 13724: **FAIL/SKIP** ($q = 6861 = 3 \cdot 2287$ NOT prime; $\equiv 1 \pmod 4$ but composite)
- Kronecker for 13724: **FAIL/SKIP** ($4 \cdot 3431$; $3431 \equiv 3 \pmod 4$ NOT Hadamard-admissible; $3431 = 47 \cdot 73$ (odd composite))
- Skip: Miyamoto digression this wave

**SPK: IN SPK via Paley-I**.
**Gap 2052:** OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_13724.py`

**Results.** overall_pass=True; units 597/597; row blocks 108/108 (block size 128); wall 2323.9573s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-07T11:26:16Z (~7:26AM ET)

## Arithmetic

- Target $n = 13724 = q + 1$ with $q = 13723$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 6861 = 3 \cdot 2287$ NOT prime; $\equiv 1 \pmod 4$ but composite — FAIL/SKIP.
- Kronecker: $13724 = 4 \cdot 3431$; $3431 \equiv 3 \pmod 4$ NOT Hadamard-admissible; $3431 = 47 \cdot 73$ (odd composite) — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

## Constructions (verified)

| Object | Method | max\|HHˆT-nI\| | Wall (s) |
|---|---|---|---|
| H_13724 Paley-I | Paley-I q=13723 | 0 | 2287.522 |
| H_13724 Paley-II | FAIL/SKIP (q=6861=3*2287 NOT prime; ===1 mod 4 but composite) | – | – |

| Object | Method | max\|HHˆT-nI\| | Wall (s) |
|---|---|---|---|
| H_13724 Kronecker | FAIL/SKIP (*43431; 3431===3 mod 4 NOT Hadamard-admissible; 3431=4773* (odd composite)) | – | – |

## SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; …; **NOT** 8948/8644/8516/…) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

| Quantity | Value |
|---|---|
| 12488 in SPK | **True** (Paley-I q=13723) |
| Gap 2052 | **OPEN** |
| clay_closed | false |
| GATE-04 | LOCKED |

## Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims

- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

## Next optional

q=13751->n=13752 Paley-I DIRECT (prime ===3 mod 4; 13729 prime ===1 mod 4 skipped; no prime ===3 mod 4 in 13724..13750; Paley-II q=6875=5?*11 NOT prime; ===3 mod 4 needs ===1 -> FAIL/SKIP; Kronecker 13752=4*3438; 3438=2$3^{2191}$ ===2 mod 4 NOT Hadamard-admissible -> FAIL/SKIP; do NOT invent H_ ===8 mod 16 = 13688/13592/13464/13400 shape) (or gap 2052 / Track B)